

Syllabus

CSCI 6461-11 Computer System Architecture - Fall 2025

Tuesday 12:45—3:15pm, Tompkins 201

Instructor: Prof. Bhagi Narahari

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Office hours: Thursday 4-6pm (in SEH5800). Other times by appointment.

Pre-requisites: Programming and Data structures, Proficiency in C programming language, Discrete Math, and a prior course in Computer organization.

Learning Outcomes

As a result of completing this course, students will be able to:

- Describe the functional architecture of various computer systems and the advances in computer organization that lead to increased performance.
- Understand and identify the components, and their interaction, in a processor
- Improve performance of programs through code optimization techniques
- Describe memory organization and analyze memory, and cache memory in particular, organizations.
- Describe parallelism (instruction level, ILP, data parallelism, and task parallelism) and analyze effectiveness, challenges, and limits of ILP processors, GPUs and Multi-core processors.
- Analyze published materials describing specific computer system architectures to determine tradeoffs made in their organizational structure and functions with respect to performance and efficiency.
- Perform comparative analyses of computer architecture alternatives based on cost, performance, power, reliability and efficiency constraints.

Workload

The course will be taught using a combination of in-person lectures and notes (and some recorded videos). As a 3 credit course, it will require a minimum of 2.5 hours per week of direct instruction and minimum of 6 hours of independent learning. The hours may be higher if you have not had a comprehensive Computer architecture course during your undergraduate program. Over the course of the semester, your independent learning will include watching video recordings, readings (lecture notes, textbook, and papers), and homeworks/projects (some homeworks will require multiple weeks and programming, and some may consist of in-class exercises). If lectures have to be conducted online then recordings will be made available on Blackboard.

Textbook(s):

Required: *Computer Architecture: A Quantitative Approach*, by John L. Hennessy and David A. Patterson

Publisher: Morgan Kaufmann; 6th Edition, ISBN-10: 0128119051 (also available online).

Required: Paper readings (published papers) will be assigned.

Additional reference (for ILP processors only: *Modern Processor Design: Fundamentals of Superscalar*

Processors, by Shen and Lipasti, ISBN 10:1-4786-0783-1, Waveland Press, available online, (recommended for topics on pipelining and ILP processors).

In addition to the textbooks, you will be required to learn and use SMP Cache simulator and the SEAS server to compile and test C programs.

Schedule of topics (subject to change):

Please check the academic calendar for fall break and thanksgiving breaks bulletin.gwu.edu/academic-calendar/

The chapters listed for each topic refer to chapters in the H&P textbook.

Date	Topic(s) and readings	Assignment(s) due
Week 1 (Aug 26)	Introduction – Performance metrics, design tradeoffs. Chapter 1 Course pre-requisite check and review – Computer organization.	
Week 2 (Sept 2)	Review – Computer Architecture-ISA and Assembly Language; Appendix A Review – Pipelined Processors, Appendix C	HW1
Week 3 (Sept 9)	Review –Pipeline limits- branch hazards and branch prediction. Intro to ILP. Appendix C and Chapter 3 intro.	
Week 4 (Sept 16)	Memory Hierarchy – Cache Organization Appendix B, Chapter 2	HW2
Week 5 (Sept 23)	Cache Optimizations Appendix B, Chapter 2	HW3 (HW4 – Project 1 on Cache design analysis assigned)
Week 6 (Sept 30)	Instruction level Parallelism (ILP) processors- Chapter 3, Appendix C	
Week 7 (Oct 7)	ILP Processors- Speculation, Scheduling Chapter 3, reference book	
Week 8 (Oct 14)	Mid-term Exam (Focuses on ILP processors and memory design: Chapters 1-3, Appendices A,B,C)	HW5, HW4-Project 1 due.
Week 9 (Oct 21)	Intro to Compiler and code optimization	HW4-Project 1 due

Week 10 (Oct 28)	ILP -Compiler Optimization wrapup Chapter 3, Appendix H, Notes Intro to Parallelism & parallel algorithms – Chap 4	(HW6 Code Optimization Assigned) Readings: Notes, Chap.4
Week 11 (Nov 4)	Data Parallelism -SIMD, Vector processing, GPU: Chapter 4	HW7 assigned
Week 12 (Nov 11)	GPU programming and CUDA overview – Chap 4 Intro to multiprocessors and Multi-core – Chap.5 Chapter 4	HW6 Due- must show demo to the TA. (HW8 – Project 3: multiprocessor cache design assigned)
Week 13 (Nov 18)	Multiprocessors-Cache Coherence, memory consistency. Chapter 5.	HW7 Due. Must show demo to the TA.
Week 14 (Dec.2)	Advanced topics and guest lectures: Domain Specific Architectures -Chapter 7	HW8/Project 3 Due.
	Exam 2: focuses on Code optimization, Data parallelism, multi-core, parallel algorithms, domain specific architectures and advanced topics (Chapters 3,4,5,7, App H, Notes). Requires knowledge of material from Exam 1 topics.	

Grading

The assessment for this course consists of two in-class exams and several homeworks. Upon approval of the instructor you can substitute a homework with an independent project that requires implementation and testing – if you want to explore this option, meet with the instructor no later than 4th week of class. The goal of the homeworks is to improve your learning of the concepts covered in the lectures. Some of the homeworks will be team based projects, while the others will be individual homeworks with *no collaboration of any kind permitted (this includes using outside sources to find the answers to the assignment)*. You cannot use sources (online or otherwise) or LLM tools to solve the homework questions – failure to follow this policy will be treated as an academic integrity violation (with a minimum score of 0 on the work product and possibly additional penalties).

There are **no late submissions** allowed in this course. However, you are allowed to submit exactly ONE late assignment (homework or project) within 48 hours of the deadline during the entire course - and you will incur a 10% penalty for this one late submission. The course grade will be based on your weighted average and it will be scaled/curved as a percentage of the highest score (i.e, the highest score is scaled to 100). The approximate grade ranges are: (A,A-)= 90-100. (B to B+)=81—89, (C+ to B)= 71-80, (C- to C)= 60-70.

- Exams account for 75% of the grade.
- Homeworks account for 25% of the grade.

University policies

Academic Integrity Code

Academic integrity is an essential part of the educational process, and all members of the GW community take these matters very seriously. As the instructor of record for this course, my role is to provide clear expectations and uphold them in all assessments. Violations of academic integrity occur when students fail to cite research sources properly, engage in unauthorized collaboration, falsify data, and otherwise violate the [Code of Academic Integrity](#). Violation of these policies will result in a minimum punishment of a zero on the assigned work, and multiple violations will result in recommending one grade lower than what you earn. If you have any questions about whether particular academic practices or resources are permitted, you should ask me for clarification. If you are reported for an academic integrity violation, you should contact Conflict Education and Student Accountability (CESA), formerly known as Student Rights and Responsibilities (SRR), to learn more about your rights and options in the process. Consequences can range from failure of assignment to expulsion from the University and may include a transcript notation. For more information, refer to the CESA website at students.gwu.edu/code-academic-integrity or contact CESA by email cesa@gwu.edu or phone 202-994-6757.

University policy on observance of religious holidays

Students must notify faculty during the first week of the semester in which they are enrolled in the course, or as early as possible, but no later than three weeks prior to the absence, of their intention to be absent from class on their day(s) of religious observance. If the holiday falls within the first three weeks of class, the student must inform faculty in the first week of the semester. For details and policy, see provost.gwu.edu/policies-procedures-and-guidelines.

Use of Electronic Course Materials and Class Recordings

Students are encouraged to use electronic course materials, including recorded class sessions, for private personal use in connection with their academic program of study. Electronic course materials and recorded class sessions should not be shared or used for non-course related purposes unless express permission has been granted by the instructor. Students who impermissibly share any electronic course materials are subject to discipline under the Student Code of Conduct. Contact the instructor if you have questions regarding what constitutes permissible or impermissible use of electronic course materials and/or recorded class sessions. Contact Disability Support Services at disabilitysupport.gwu.edu if you have questions or need assistance in accessing electronic course materials.

Academic support

Academic Commons

[Academic Commons](#) is the central location for academic support resources for GW students. To schedule a peer tutoring session for a variety of courses visit go.gwu.edu/tutoring. Visit academiccommons.gwu.edu for study skills tips, finding help with research, and connecting with other campus resources. For questions email academiccommons@gwu.edu.

Support for students in and outside the classroom

Disability Support Services (DSS) 202-994-8250

Any student who may need an accommodation based on the potential impact of a disability should contact Disability Support Services at disabilitysupport.gwu.edu to establish eligibility and to coordinate reasonable accommodations.

Student Health Center 202-994-5300, 24/7

The Student Health Center (SHC) offers [medical](#), [counseling/psychological](#), and [psychiatric](#) services to GW students. More information about the SHC is available at healthcenter.gwu.edu. Students experiencing a medical or mental health emergency on campus should contact GW Emergency Services at 202-994-6111, or off campus at 911.

GW Campus Emergency Information

GW Emergency Services: 202-994-6111

For situation-specific instructions, refer to [GW's Emergency Procedures guide](#).

GW Alert

GW Alert is an emergency notification system that sends alerts to the GW community. GW requests students, faculty, and staff maintain current contact information by logging on to alert.gwu.edu. Alerts are sent via email, text, social media, and other means, including the Guardian app. The Guardian app is a safety app that allows you to communicate quickly with GW Emergency Services, 911, and other resources. Learn more at safety.gwu.edu.

Protective Actions

GW prescribes four protective actions that can be issued by university officials depending on the type of emergency. All GW community members are expected to follow directions according to the specified protective action. The protective actions are Shelter, Evacuate, Secure, and Lockdown (details below). Learn more at safety.gwu.edu/gw-standard-emergency-statutes.